一.

$m_X(t) = EX(t) = aE\cos(\theta+\omega t) = 0.$

$R_X(t,t+\tau) = EX(t)X(t+\tau)$

$= a^2 E\cos(\theta+\omega t)\cos(\theta+\omega t+\omega\tau)$

$= \frac{a^2}{2}\cdot E[\cos(2\theta+2\omega t+\omega\tau) + \cos\omega\tau]$

$= \frac{a^2}{2}\cos\omega\tau.$

$D_X(t) = EX^2(t) - [EX(t)]^2$

$= a^2 E\cos^2(\theta+\omega t)$

$= \frac{a^2}{2}$

二.

1. $m_X(t) = EX(t) = \cos\omega t\, EA + \sin\omega t\, EB = 0.$

$D_X(t) = EX^2(t) - [EX(t)]^2 = E[A^2\cos^2(\omega t) + B^2\sin^2(\omega t) + 2AB\sin(\omega t)\cos(\omega t)]$

$= \cos^2(\omega t)EA^2 + \sin^2(\omega t)EB^2$

$= \sigma^2$

$R_X(t,t+\tau) = E[X(t)X(t+\tau)]$

$= E[A^2\cos(\omega t)\cos(\omega t+\omega\tau) + B^2\sin(\omega t)\sin(\omega t+\omega\tau) + AB\sin\omega t\cos(\omega t+\omega\tau) + AB\cos\omega t\sin(\omega t+\omega\tau)]$

$= \sigma^2\cdot[\cos\omega t\cos(\omega t+\omega\tau) + \sin\omega t\sin(\omega t+\omega\tau)]$

$= \sigma^2\cos\omega\tau$

2. $X(t) \sim (0,\sigma^2)$ $f_X(t) = \frac{1}{\sqrt{2\pi}\sigma}e^{-\frac{x^2}{2\sigma^2}}$

3. $R_X(1,2) = \sigma^2\cos\omega = R_X(2,1)$

$R_X(1,3) = R_X(3,1) = \sigma^2\cos 2\omega$

$R_X(2,3) = R_X(3,2) = \sigma^2\cos\omega.$

协差矩阵:

$$\begin{bmatrix} \sigma^2 & \sigma^2\cos\omega & \sigma^2\cos 2\omega \\ \sigma^2\cos\omega & \sigma^2 & \sigma^2\cos\omega \\ \sigma^2\cos 2\omega & \sigma^2\cos\omega & \sigma^2 \end{bmatrix}$$

三.

1. $m_X(t) = \lambda_1 t$

$D_X(t) = \lambda_1 t$

$R_X(t,t+\tau) = \lambda_1^2 t(t+\tau) + \lambda_1 t$

$R_X(t,t+\tau) = E[X(t)X(t+\tau)]$

$= E[X(t)[X(t+\tau) - X(t)] + X^2(t)]$

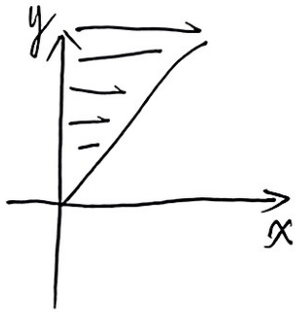
$= EX(t)\,E[X(t+\tau) - X(t)] + EX^2(t)$

$= \lambda_1 t\cdot\lambda_1\tau + \lambda_1^2 t^2 + \lambda_1 t$

$= \lambda_1^2 t(t+\tau) + \lambda_1 t$

2. $f_{W_x(1)}(x) = \lambda_1 e^{-\lambda_1 x}$

$f_{W_y(1)}(y) = \lambda_2 e^{-\lambda_2 y}.$



$P\{W_x^{(1)} < W_y^{(1)}\}$

$= \int_0^{+\infty}\int_0^{y}\lambda_1 e^{-\lambda_1 x}\lambda_2 e^{-\lambda_2 y}\,dx\,dy.$

$= \int_0^{+\infty}(1-e^{-\lambda_1 y})\lambda_2 e^{-\lambda_2 y}\,dy$

$= \frac{\lambda_1}{\lambda_1+\lambda_2}$

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四.

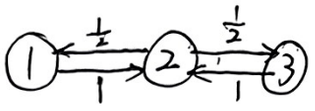
1.

$$P=\begin{bmatrix}0&1&0\\ \frac12&0&\frac12\\ 0&1&0\end{bmatrix}\qquad P^2=\begin{bmatrix}\frac12&0&\frac12\\ 0&1&0\\ \frac12&0&\frac12\end{bmatrix}$$

$$P^3=\begin{bmatrix}0&1&0\\ \frac12&0&\frac12\\ 0&1&0\end{bmatrix}=P$$

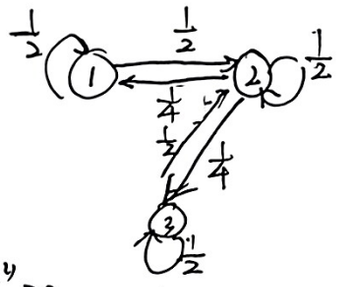
不存在 n，使得 $\forall P_{ij}^{(n)}\in P^n$ $\quad P_{ij}^{(n)}>0$.

不是遍历的



2.

$$P^2=\begin{bmatrix}\frac58&\frac12&\frac18\\ \frac14&\frac12&\frac14\\ \frac18&\frac12&\frac38\end{bmatrix}$$



$\forall P_{ij}^{(2)}\in P^2$ $\quad P_{ij}^{(2)}>0$. 又 $d=1$

故为遍历链.

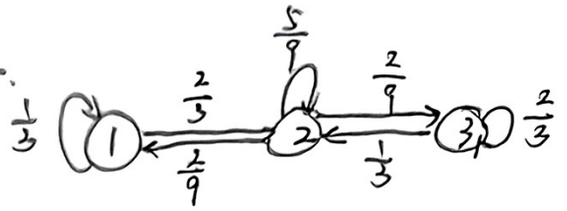
设其平稳分布为 $\pi=[\pi_1,\pi_2,\pi_3]$

$$\pi P=\pi.$$

$$\Rightarrow\begin{cases}\pi_1=\frac12\pi_1+\frac14\pi_2\\ \pi_2=\frac12\pi_1+\frac12\pi_2+\frac12\pi_3\\ \pi_3=\frac14\pi_2+\frac12\pi_3\\ \pi_1+\pi_2+\pi_3=1\end{cases}$$

$$\Rightarrow\pi=\left[\frac14,\frac12,\frac14\right].$$

五.



易知：可以链常返且 $d=1$.

$$P^2=\begin{bmatrix}\frac{7}{27}&\frac{16}{27}&\frac{4}{27}\\ \frac{16}{81}&\frac{43}{81}&\frac{22}{81}\\ \frac{2}{27}&\frac{11}{27}&\frac{14}{27}\end{bmatrix}.$$

$\forall P_{ij}^{(2)}>0$. 易知其不可约.

故遍历.

2.

$$[1,0,0]\begin{bmatrix}\frac13&\frac23&0\\ \frac29&\frac59&\frac29\\ 0&\frac13&\frac23\end{bmatrix}=\left[\frac13,\frac23,0\right].$$

$X(n=1)$	0	1	2
$P\{X(1)\}$	$\frac13$	$\frac23$	0

3. 设平稳分布为 $\pi=[\pi_1,\pi_2,\pi_3]$.

$$\pi P=\pi$$

$$\begin{cases}\pi_1=\frac13\pi_1+\frac29\pi_2\\ \pi_2=\frac23\pi_1+\frac59\pi_2+\frac13\pi_3\\ \pi_3=\frac29\pi_2+\frac23\pi_3\\ \pi_1+\pi_2+\pi_3=1\end{cases}$$

$$\Rightarrow\pi=\left[\frac16,\frac12,\frac13\right]$$

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六.

$$m_X(t) = EX(t) = E\sin(2\pi t+\theta) = \int_0^1 \sin(2\pi t+\theta)\,d\theta$$

$$= \sin(2\pi t) \neq C.$$

$X(t)$ 不具有平稳性.

七.

(1) $m_X(t) = \frac{1}{2}\times 1 + \frac{1}{2}\times(-1) = 0$.

$$R_X(t,t+\tau) = E[X(t)X(t+\tau)]$$

$$= \sum_{k=1}^{+\infty} \frac{e^{-\lambda\tau}}{(2k)!}(\lambda\tau)^{2k} - \sum_{k=1}^{+\infty} \frac{e^{-\lambda\tau}}{(2k-1)!}(\lambda\tau)^{(2k-1)}$$

$$= \sum_{k=1}^{+\infty} \frac{e^{-\lambda\tau}}{k!}(-\lambda\tau)^k$$

$$= e^{-2\lambda\tau}$$ 只与 τ 有关

$\tau<0$ 时, $R_X(t,t+\tau) = e^{2\lambda\tau}$ $\Rightarrow R_X(t,t+\tau) = e^{-2\lambda|\tau|}$

$EX^2(t) = \frac{1}{2}\times 1 + \frac{1}{2}\times(-1)^2 = 1 < +\infty$.

$X(t)$ 为宽平稳过程.

(2) $$S_X(\omega) = \int_{-\infty}^{+\infty} R_X(t,t+\tau)\,e^{-j\omega\tau}\,d\tau$$

$$= \frac{4\lambda}{\omega^2+4\lambda^2}$$

八.

$$S_X(\omega) = \frac{2}{\omega^4+5\omega^2+4} = \frac{1}{3}\frac{2}{\omega^2+1} - \frac{1}{6}\frac{4}{\omega^2+4}$$

$$\Rightarrow R_X(\tau) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} S_X(\omega)\,e^{j\tau\omega}\,d\omega$$

$$= \frac{1}{3}e^{-|\tau|} - \frac{1}{6}e^{-2|\tau|}$$

$$\Psi^2 = R_X(0) = \frac{1}{6}$$

九.

1. $$R_X(\tau) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} S_X(\omega)\,e^{j\omega\tau}\,d\omega = e^{-|\tau|}$$

2. $m_X(t) = 0$

$$EX^2(t) = R_X(0) = 1 = D_X(t)$$

$$\Rightarrow X(t) \sim N(0,1).$$

$$f_X(t) = \frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}$$

十

$$R_X(\tau) = e^{-|\tau|}$$

$$S_X(\omega) = \frac{2}{\omega^2+1}$$

$$H(j\omega) = \frac{\frac{1}{j\omega C}}{R+\frac{1}{j\omega C}} = \frac{1}{j\omega RC+1}$$

$$S_Y(\omega) = S_X(\omega)\cdot|H(j\omega)|^2 = \frac{2}{\omega^2+1}\cdot\frac{1}{1+\omega^2R^2C^2}$$

$$= \frac{2}{\omega^2+1}\cdot\frac{\beta^2}{\omega^2+\beta^2}$$

$$= \frac{\beta^2}{\beta^2-1}\frac{2}{\omega^2+1} - \frac{\beta^2}{\beta^2-1}\frac{2\beta}{\omega^2+\beta^2}$$

$$R_Y(\tau) = \frac{\beta^2}{\beta^2-1}e^{-|\tau|} - \frac{\beta}{\beta^2-1}e^{-\beta|\tau|}$$

$$m_Y(t) = m_X(t)\int_{-\infty}^{+\infty} h(u)\,du = 0.$$

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